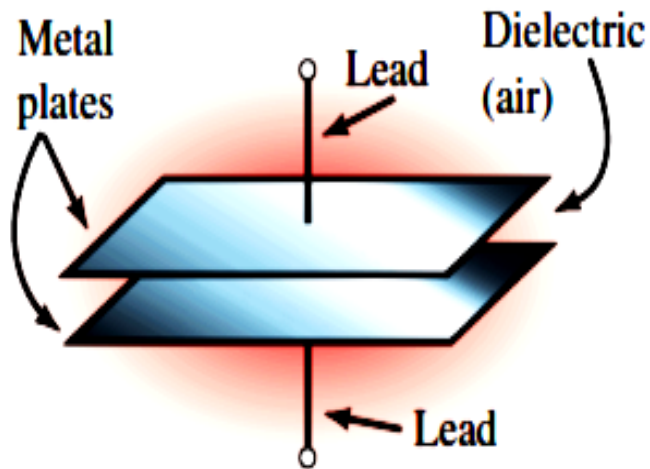


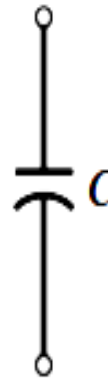
CAPACITORS *AND* *INDUCTORS*

4.1 CAPACITANCE

- A capacitor consists of **two conductors separated by an insulator**.
- One of its basic forms is the parallel-plate capacitor shown in Figure 4–1.
- It consists of two metal plates separated by **a non-conducting material (i.e., an insulator) called a dielectric**.
- The dielectric may be **air, oil, mica, plastic, ceramic, or other** suitable insulating material.



(a) Basic construction



(b) Symbol

FIGURE 4–1
Parallel-plate capacitor.

Cont'd

3

- Since the **plates** of the capacitor are metal, they contain **huge numbers of free electrons**.
- In their **normal state**, however, they are **uncharged**, that is, there is no excess or deficiency of electrons on either plate.
- **If a dc source** is now connected (Figure 4–2), **electrons are pulled from the top** plate by the positive potential of the battery and the same number **deposited on the bottom** plate.
- This leaves the top plate with a deficiency of electrons and the bottom plate with an excess
- In this state, the capacitor is **said to be charged**. If the amount of charge transferred during this process is Q coulombs, we say that the capacitor has a charge of Q .

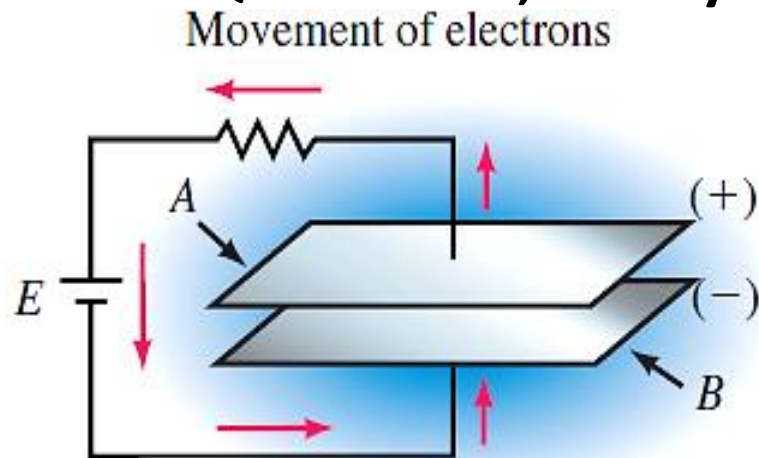


Figure 4–2

Cont'd

4

- **If we now disconnect the source** (Figure 10–3), the excess electrons that were moved to the bottom plate remain trapped as they have no way to return to the top plate.
- The capacitor therefore **remains charged** even though no source is present. Because of this, we say that **a capacitor can store charge**.
- Capacitors with little leakage can hold their charge for a considerable time.

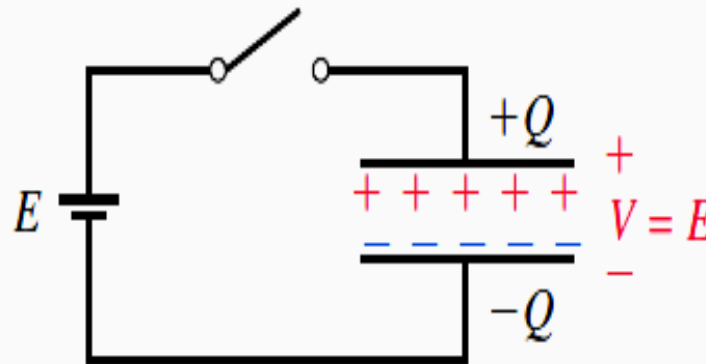


Figure 4-3

Capacitor after charging.

When the source is disconnected, electrons are trapped on the bottom plate. Thus, charge is stored.

DEFINITION OF CAPACITANCE

5

- The amount of charge Q that a capacitor can store depends on the applied voltage.
- **Experiments** show that for a given capacitor, **Q is proportional to voltage.**
- The **amount of charge that a capacitor can store per unit of voltage across its plates is its capacitance, designated by C .**
- That is, capacitance is a **measure of a capacitor's ability to store charge**, can be expressed by the following formula:

$$C = Q/V \dots\dots\dots(4.1)$$

- By rearranging the terms in Equation 4-1, you can obtain two other formulas.

$$Q = VC \dots\dots\dots(4.2)$$

and

$$V = Q/C \dots\dots\dots(4.3)$$

Cont'd

6

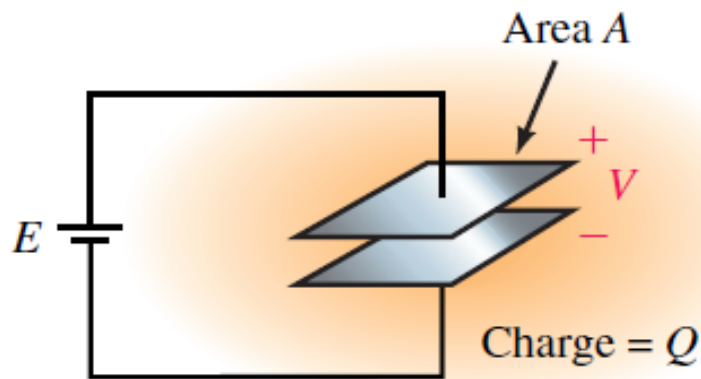
- The **SI unit** of Capacitance is **farad (F)**. Recall that the coulomb (C) is the unit of electrical charge.
- Therefore, by definition, the capacitance of a capacitor is one farad if it stores one coulomb of charge when the voltage across its terminals is one volt. (**1Farad= 1Coulomb/Volt**)
- The farad, however, is a very large unit. Most practical capacitors range in size of **pF or μF** .
- The larger the value of C, the more charge that the capacitor can hold for a given voltage.

EXAMPLE

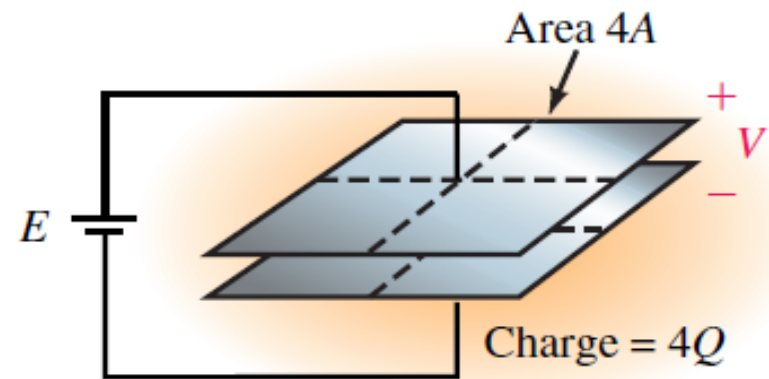
- a) How much charge is stored on a 10-mF capacitor when it is connected to a 24-volt source?
- b) A certain capacitor stores 50 micro coulombs ($50 \mu\text{C}$) with 10 V across its plates. What is its capacitance in units of microfarads?
- c) Determine the voltage across a 1000 pF capacitor that is storing 20 micro coulombs ($20 \mu\text{C}$) of charge.

A) EFFECT OF AREA (A)

- As shown by Equation 4–1, **capacitance is directly proportional to charge**.
- This means that the more charge you can put on a capacitor's plates for a given voltage, the greater will be its capacitance. Consider Figure 4–4.
- The capacitor of (b) has four times the area of (a). Since it has the same **number of free electrons per unit area**, it has four times the total charge and hence four times the capacitance.
- *This turns out to be true in general, that is, capacitance is **directly proportional to plate area**.*



(a) Capacitor with area A
and charge Q



(b) Plates with four times the area
have four times the charge and
therefore, four times the capacitance

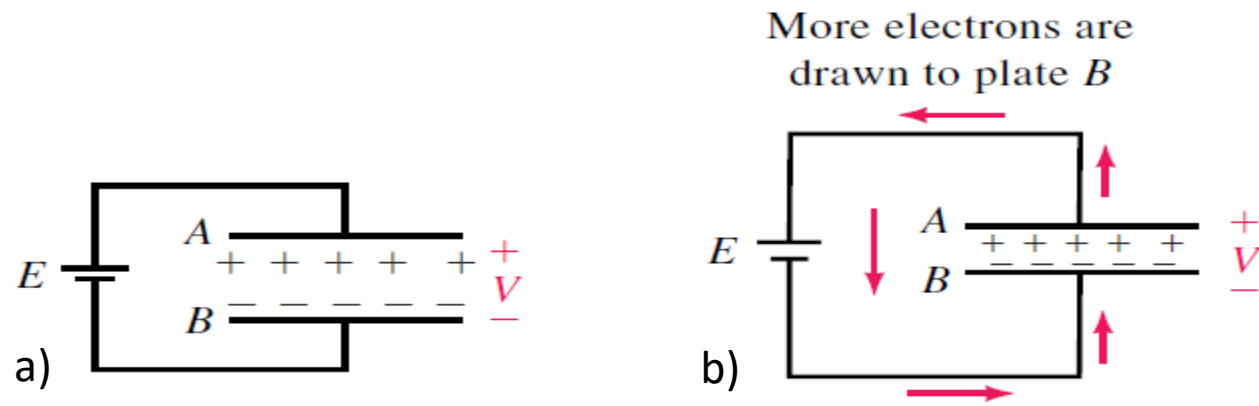
Figure 4–4

B) EFFECT OF SPACING

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- Now consider Figure 4–5. Since the top plate has a deficiency of electrons and the bottom plate an excess, a force of attraction exists across the gap.
- For a fixed spacing as in (a), the charges are in equilibrium. Now move the plates closer together as in (b).
- As spacing decreases, the force of attraction increases, pulling more electrons from within the material of plate *B* to its top surface.
- The charge on the plates therefore increases and hence, according to Equation 4–1($C = Q/V$), so does the capacitance.
- We therefore conclude that decreasing spacing increases capacitance, and vice versa. In fact, as we will show later, *capacitance is inversely proportional to plate spacing*.

FIGURE 4–5 Decreasing spacing increases capacitance.



C) EFFECT OF DIELECTRIC

- If a potential difference of V volts is applied across the two plates separated by a distance of d , the **electric field strength between the plates** is determined by

$$E = V/d \dots\dots\dots(4.4)$$

where E = electric field strength(in V/m)

V = potential difference(in V)

d = separation(in m)

- Many values of capacitance can be obtained for the same set of parallel plates by the addition of certain insulating materials between the plates.
- **The purpose of the dielectric, therefore, is to create an electric field to oppose the electric field set up by free charges on the parallel plates.**
- For this reason, the insulating material is referred to as **a dielectric, di for “opposing” and electric for “electric field.”**

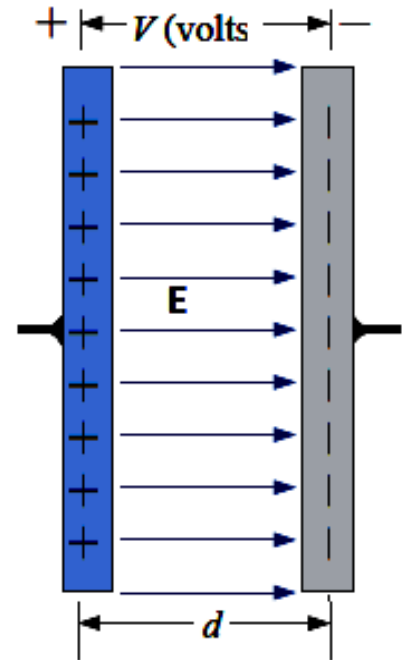


Figure 4.6

- In either case—**with or without the dielectric**—if the **potential across the plates is kept constant** and the **distance between the plates is fixed**, the **net electric field within the plates must remain the same**, as determined by the equation $E = V/d$.
- Therefore, the **net electric field between the plates would decrease with insertion of the dielectric** for a fixed amount of free charge on the plates.
- To compensate and keep the net electric field equal to the value determined by V and d , more charge must be deposited on the plates.
- This additional charge for the same potential across the plates **increases the capacitance** (since $C = Q/V$).
- For different dielectric materials between the same two parallel plates, different amounts of charge will be deposited on the plates.

- But, $\Psi = Q$, (where ψ = electric flux lines) so the dielectric is also determining the number of flux lines between the two plates and consequently the flux density ($D = \psi/A$) since A is fixed.
- By definition, the ratio of the flux density(D) to the electric field intensity(E) in the dielectric is called the permittivity of the dielectric(ϵ).

$$\epsilon = D/E \quad [\text{F/m}] \dots \dots \dots (4.5)$$

where ϵ = permittivity of the dielectric;

D = Electric flux density and E = electric field strength

- Permittivity(ϵ) is a measure of how easily the dielectric will “permit” the establishment of flux lines within the dielectric.
- The greater its value, the greater the amount of charge deposited on the plates, and, consequently, the greater the flux density for a fixed area.

- For a vacuum, the value of ϵ (denoted by ϵ_0) is $\epsilon_0 = 8.85 \times 10^{-12}$ F/m.
- The ratio of the permittivity of any dielectric to that of a vacuum is called the relative permittivity, ϵ_r .
- It simply compares the permittivity of the dielectric to that of air and in equation form,

$$\epsilon_r = \epsilon / \epsilon_0 \dots\dots\dots(4.6)$$

- The value of ϵ for any material, therefore, is

$$\epsilon = \epsilon_r \epsilon_0 \dots\dots\dots(4.7)$$

- Note that ϵ_r is a dimensionless quantity.
- The relative permittivity for various dielectric materials is provided in Table 4.1

TABLE 4-1 Relative Dielectric Constants (Also Called Relative Permittivities)

Material	ϵ_r (Nominal Values)
Vacuum	1
Air	1.0006
Ceramic	30–7500
Mica	5.5
Mylar	3
Oil	4
Paper (dry)	2.2
Polystyrene	2.6
Teflon	2.1

- Substituting for D and $\mathcal{E}$ in Eq. (4.5), we have

$$\epsilon = \frac{D}{\mathcal{E}} = \frac{\psi/A}{V/d} = \frac{Q/A}{V/d} = \frac{Qd}{VA}$$

- But $C = Q/V$ and, therefore, $\epsilon = Cd/A$, from this, we can drive C as follow:

$$C = \epsilon \frac{A}{d} \quad (\text{F}) \quad \text{..... (4.8)}$$

- Since, $\epsilon = \epsilon_r \epsilon_0$, then C can be written as :

$$C = \epsilon_0 \epsilon_r \frac{A}{d} \quad \text{.....(4.9)}$$

where A is the area of the plates, d is the distance between the plates, and ϵ_r is the relative permittivity.

- From the above observations, we see that capacitance is directly proportional to plate area, inversely proportional to plate separation, and dependent on the dielectric permittivity.

- Consider, again, Equation 4–9: $C = \epsilon A/d = \epsilon_r \epsilon_o A/d$.
- Note that, $C_o = \epsilon_o A/d$ is the capacitance of a vacuum- (or air-) dielectric capacitor. Denote it by C_o .
- Then, for any other dielectric,

$$C = \epsilon_r C_o \quad \dots\dots\dots(4.10)$$

- The electric field intensity(E) between the plates can be written in terms of the permittivity ϵ , the charge Q , and the surface area A of the plates as follow:

- we know that $C = Q/V$ and $C = \epsilon A/d$

$$\text{then, } Q/V = \epsilon A/d$$

$$V/d = Q/\epsilon A$$

$$\text{but } V/d = E = Q/\epsilon A$$

$$\text{therefore, } E = \frac{Q}{\epsilon A} \quad \text{Where , } E \text{ is electric field intensity}$$

Example

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- 1) Compute the capacitance of a parallel-plate capacitor with plates 10 cm by 20 cm, separation of 5 mm, and
 - a) an air dielectric,
 - b) a ceramic dielectric with permittivity of 7500.
- 2) A capacitor with plates 7.5 cm by 8 cm and plate separation of 0.1 mm has an oil dielectric:
 - a) Compute its capacitance;
 - b) If the charge on this capacitor is $0.424 \mu\text{C}$, what is the voltage across its plates?
- 3) For the capacitor of Figure 4.7 below
 - a) Determine the capacitance.
 - b) Determine the electric field strength between the plates if 450 V are applied across the plates.
 - c) Find the resulting charge on each plate.

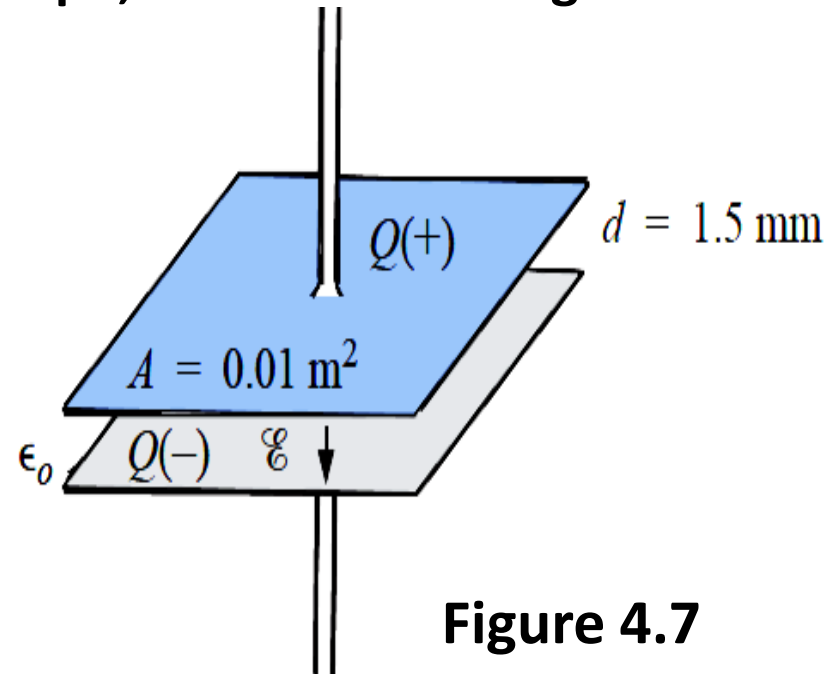


Figure 4.7

- A capacitor stores energy in the form of an electric field that is established by the opposite charges stored on the two plates.
- The **greater the forces** between the charges on the plates of a capacitor, the **more energy** is stored.
- Also, from Equation $Q = VC$, the amount of **charge** stored is **directly related to the voltage** as well as to the **capacitance**.
- Therefore, the **amount of energy stored is directly proportional to the capacitance** because the more charge stored, the greater the force.
- The formula for the energy stored by a capacitor is

$$W_C = \frac{1}{2}CV^2 \quad \text{.....(4.11)}$$

- In terms of Q and C , since $V=Q/C$ and substituting this,

$$W_C = \frac{Q^2}{2C} \quad \text{.....(4.12)}$$

A) CAPACITORS IN SERIES

- Capacitors, like resistors, can be placed in series as figure 4.8.

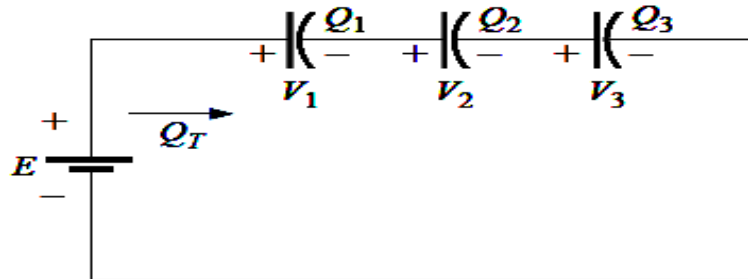


Figure 4.8
Series capacitors.

- For capacitors in series, the charge is the same on each capacitor:

$$Q_T = Q_1 = Q_2 = Q_3 \quad \text{.....(4.13)}$$

- Applying Kirchhoff's voltage law around the closed loop gives,

$$E = V_1 + V_2 + V_3$$

- However, $V = \frac{Q}{C}$, so that $\frac{Q_T}{C_T} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2} + \frac{Q_3}{C_3}$

- Using Eq. (4.12) and dividing both sides by Q yields:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad \text{.....(4.14)}$$

- The total capacitance of two capacitors in series is:

$$C_T = \frac{C_1 C_2}{C_1 + C_2} \dots\dots\dots(4.15)$$

- The voltage across each capacitor of Fig. 4.8 can be found by first recognizing that

$$Q_T = Q_1 \text{ or } C_1 V_1 = C_T V_T$$

- Solving for V1:

$$V_1 = \left(\frac{C_T}{C_1} \right) V_T$$

- This type of relationship holds for all capacitors. Thus,

$$V_x = \left(\frac{C_T}{C_x} \right) V_T \dots\dots\dots(4.16)$$

- A similar equation will result for each capacitor of the network.

$$V_1 = \left(\frac{C_2}{C_1} \right) V_2, \quad V_1 = \left(\frac{C_3}{C_1} \right) V_3, \quad V_2 = \left(\frac{C_3}{C_2} \right) V_3, \quad \text{etc.}$$

B) CAPACITORS IN PARALLEL

- Capacitors, can be placed in parallel as figure 4.9.

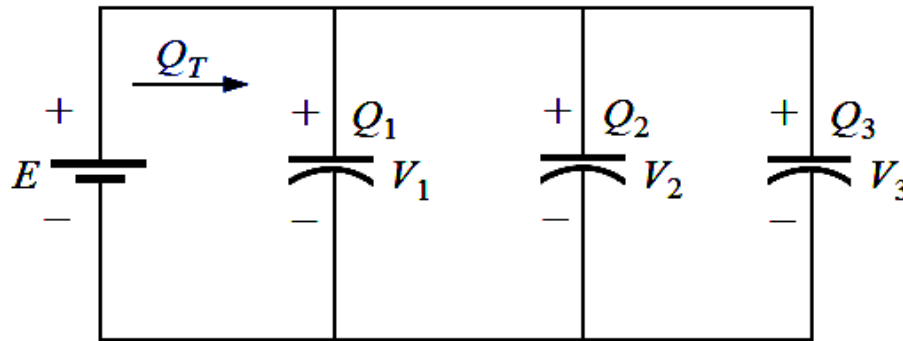


Fig. 4.9
Parallel capacitors

- For capacitors in parallel, as shown in Fig. 4.9, the voltage is the same across each capacitor, and the total charge is the sum of that on each capacitor:

$$Q_T = Q_1 + Q_2 + Q_3 \quad \text{.....(4.17)}$$

- However, $Q = CV$
- Therefore, $C_T E = C_1 V_1 + C_2 V_2 + C_3 V_3$
- But, $E = V_1 = V_2 = V_3$ thus,

$$C_T = C_1 + C_2 + C_3 \quad \text{.....(4.18)}$$

Example

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1) For the circuit of Fig. 4.10:

- Find the total capacitance.
- Determine the charge on each plate.
- Find the voltage across each capacitor.
- Determine the energy stored by each capacitor.

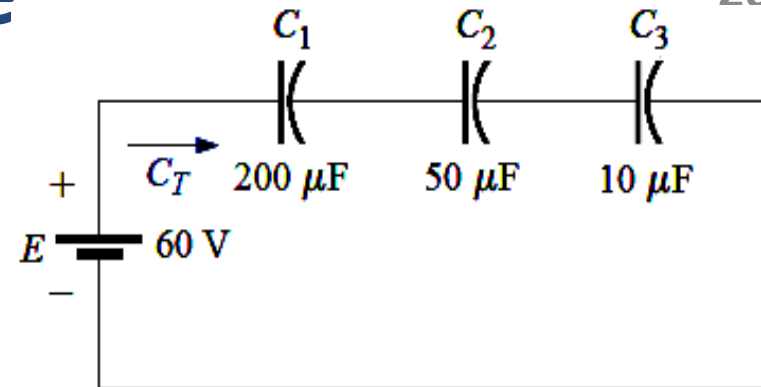


Fig 4.10

2) For the network of Fig. 4.11:

- Find the total capacitance.
- Determine the charge on each plate.
- Find the total charge.
- Determine the energy stored by each capacitor.

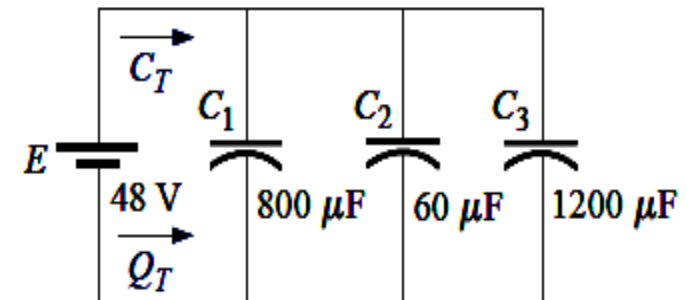


Fig 4.11

3) Find the voltage across and charge on each capacitor for the network of Fig. 4.12.

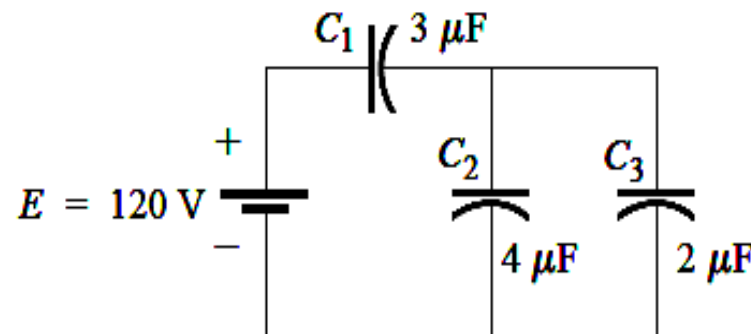


Fig 4.12

4.2 INDUCTORS

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FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

- If a **conductor is moved through a magnetic field** so that it cuts magnetic lines of flux, **a voltage will be induced** across the conductor, as shown in Fig. 4.13.
- The **greater the number of flux lines cut per unit time** (by **increasing the speed** with which the conductor passes through the field), or the **stronger the magnetic field strength** (for the same traversing speed), the **greater** will be the **induced voltage** across the conductor.
- If the **conductor is held fixed** and the **magnetic field is moved** so that its flux lines cut the conductor, the **same effect** will be produced.

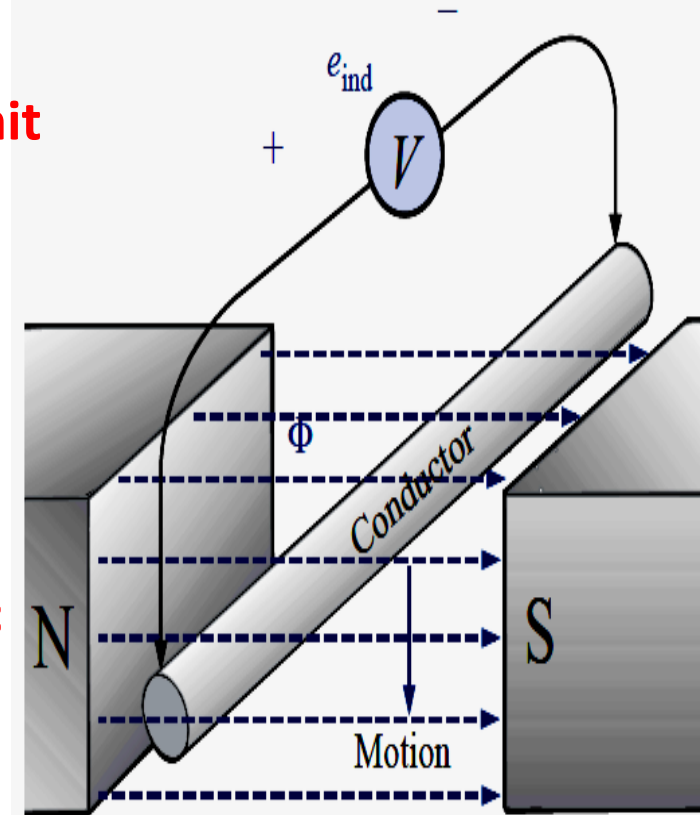


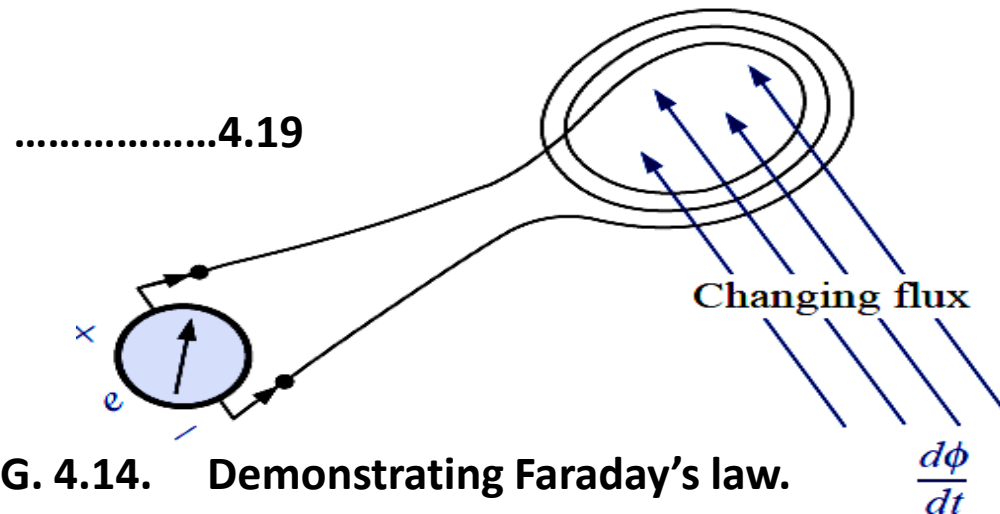
FIG. 4.13

Generating an induced voltage by moving a conductor through a magnetic field.

- **Michael Faraday** discovered the principle of electromagnetic induction in 1831.
- Faraday's two observations are stated as follows:
 - a) The amount of **voltage induced** in a coil is **directly proportional to the rate of change of the magnetic field** with respect to the coil ($d\phi/dt$).
 - b) The amount of voltage induced in a coil is **directly proportional to the number of turns** of wire in the coil (N).
- If a coil of **N turns** is placed in the region of a changing flux, as in Fig. 4.14, a voltage will be induced across the coil as determined by Faraday's law:

$$v_{\text{ind}} = N \left(\frac{d\phi}{dt} \right)$$

where N represents the number of turns of the coil and $d\phi/dt$ is the instantaneous change in flux (in webers) linking the coil.



- Lenz's law **defines the polarity or direction of the induced voltage.**
- In chapter 3, it was shown that the magnetic flux linking a coil of N turns with a current I has the distribution of Fig. 4.15.
- If the **current increases** in magnitude, the **flux** linking the coil also **increases.**
- It was shown, however, that a **changing flux** linking a coil **induces a voltage** across the coil.
- Therefore, an induced voltage is developed across the coil due to the change in current through the coil.
- In other words, **the induced effect (e_{ind})** is a **result of the increasing current** through the coil.
- This effect is an example of a general principle known as Lenz's law, which states that **an induced effect is always such as to oppose the cause that produced it.**

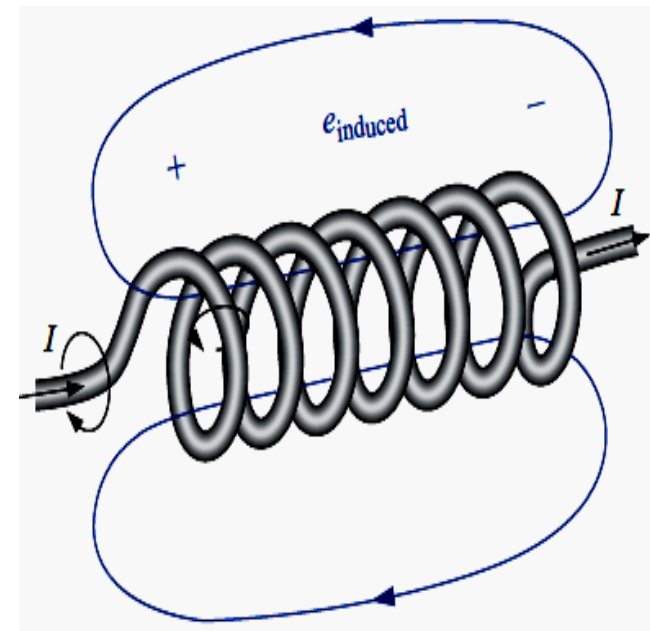


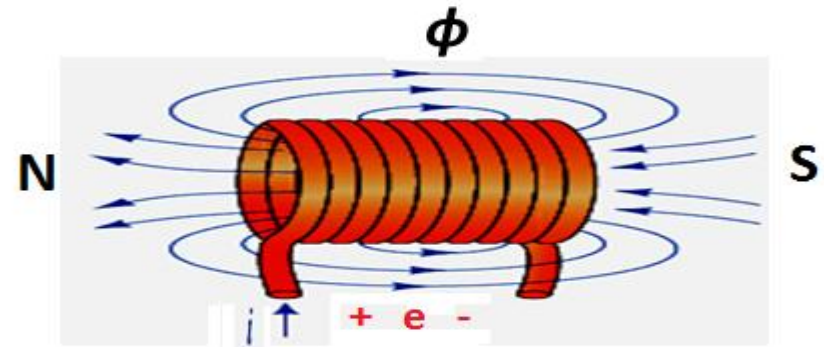
Fig. 4.15.

Demonstrating the effect of Lenz's law.

- **When a length of wire is formed into a coil**, as shown in Figure 4.16, it becomes an inductor.

Figure 4.16

A coil of wire forms an inductor.



- The terms coil and inductor are used interchangeably.
- **Current through the coil produces an electromagnetic field**, as illustrated.
- The magnetic lines of force around each loop (turn) in the winding of the coil effectively add to the lines of force around the adjoining loops, forming a strong electromagnetic field within and around the coil.
- Figure 4.17 shows a schematic symbol for an inductor.

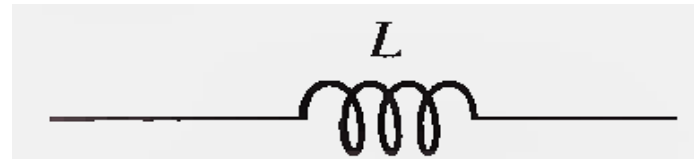


Figure 4.17

Symbol for an inductor

- When there is current through an inductor, an electromagnetic field is established.
- The changing current through an inductor produces a changing electromagnetic field around the inductor.
- In turn, the changing electromagnetic field causes an induced voltage across the coil in a direction to oppose the change in current.
- This property is called self-inductance but is usually referred to as simply inductance, symbolized by L .
- Therefore, **Inductance(L) is a measure of a coil's(N) ability to establish an induced voltage as a result of a change in its current(di/dt), and that induced voltage is in a direction to oppose the change in current.**
- The inductance (L) of a coil and the time rate of change of the current (di/dt) determine the induced voltage (V_{ind}).
- Therefore, the induced voltage can be written as:

$$v_{ind} = L \left(\frac{di}{dt} \right) \dots\dots\dots(4.20)$$

- The above formula indicates that **the greater the inductance, the greater the induced voltage.**
- Also, it shows that the **faster the coil current changes (greater di/dt), the greater the induced voltage.**
- The SI unit of Inductance is Henries (H).

ENERGY STORAGE

- An inductor stores **energy in the electromagnetic field** created by the current.
- The **energy stored** is expressed as follows:

$$W = \frac{1}{2}LI^2 \quad \dots\dots\dots(4.21)$$

Where I is current and L is inductance and W is energy.

- As you can see the energy stored is proportional to the inductance and the square of the current.

- A close approximation, in terms of physical dimensions, for the inductance of the coils of Fig. 4.17. can be found using the following equation:

$$L = \frac{N^2 \mu A}{l} \dots\dots\dots(4.22)$$

where L is the inductance in henries (H), N is the number of turns of wire, μ is the permeability in henries per meter (H/m), A is the cross-sectional area in meters squared, and l is the core length in meters (m).

- The inductance is inversely proportional to the length of the core and directly proportional to the cross-sectional area. Also, the inductance is directly related to the number of turns squared.

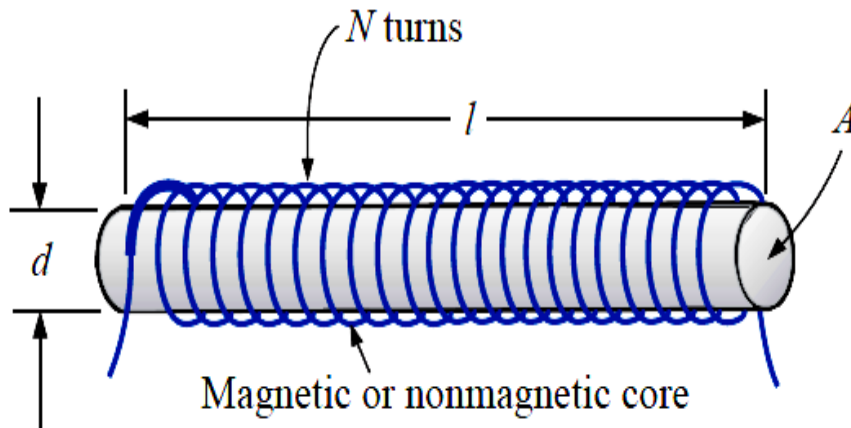


Fig. 4.17
Physical parameters of an inductor.

example

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- 1) Find the induced voltage across a coil with 500 turns located in a magnetic field that is changing at a rate of 5 Wb/s.
- 2) The current through a 15 mH inductor is changing at the rate of 500 mA/s. What is the induced voltage?
- 3) The voltage across an inductance is 250 V when its current changes at the rate of 10 mA/s. What is L?
- 4) If the voltage across a 2-mH inductance is 50 volts, how fast is the current changing?
- 5) Determine the inductance of the coil in Figure 4.18. The permeability of the core is $0.25 \times 10^{-3} \text{ H/m}$.

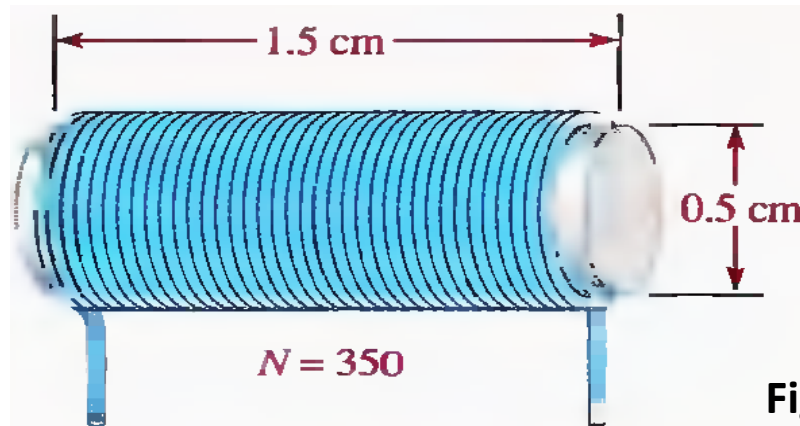


Figure 4.18.

- For inductances in series or parallel, the equivalent inductance is found by using the same rules that you used for resistance.
- For the series case (Figure 4.19) the total inductance is the sum of the individual inductances:

$$L_T = L_1 + L_2 + L_3 + \dots + L_N \quad \text{.....(4.23)}$$

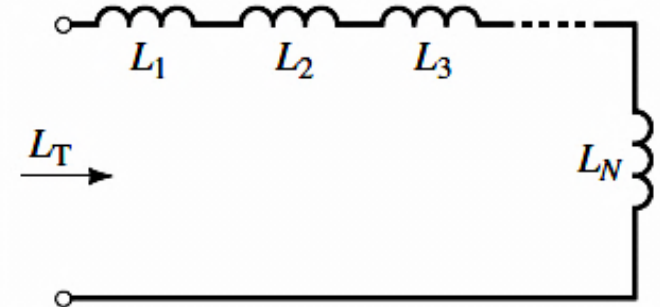


Figure 4.19

- For inductors in parallel, the total inductance is found in the same manner as the total resistance of resistors in parallel (Fig. 4.20):

$$\frac{1}{L_T} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_N} \quad \text{.....(4.24)}$$

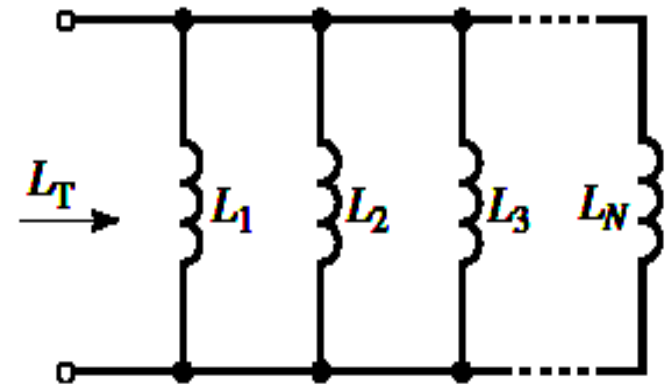


Figure 4.20

- For two inductors in parallel,

$$L_T = \frac{L_1 L_2}{L_1 + L_2} \dots\dots\dots(4.25)$$

EXAMPLE

- Find L_T for the circuit of Figure 4.21.

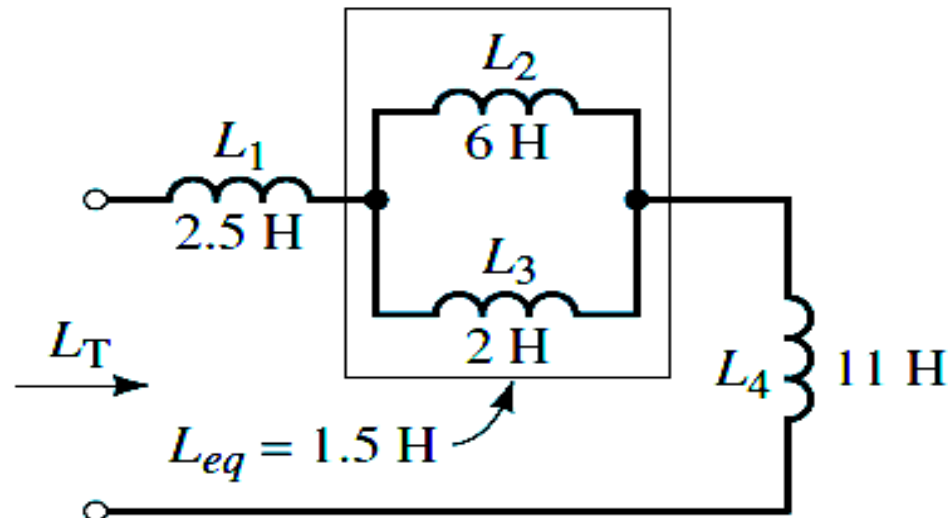


Figure 4.21

Cont'd

- 2) For Figure 4.22 , $L_T = 2.25 \text{ H}$. Determine L_x

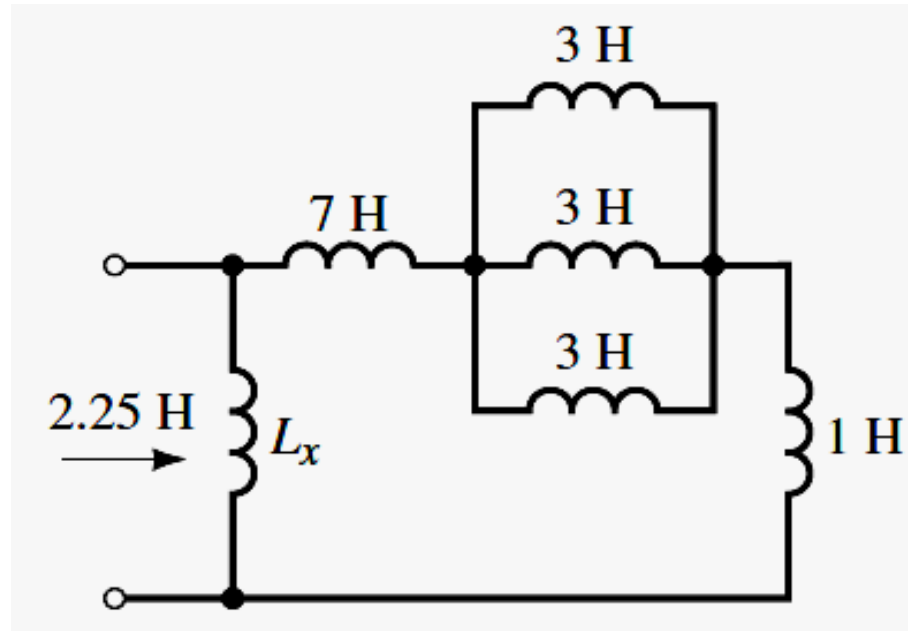


Figure 4.22

- 3) For Figure 4.19, the current is the same in each inductance and $V_1 = L_1 di/dt$, $V_2 = L_2 di/dt$, and so on. Apply KVL and show that

$$L_T = L_1 + L_2 + L_3 + \dots + L_N$$

Cont'd

4) For Figure 4.23, determine L_x .

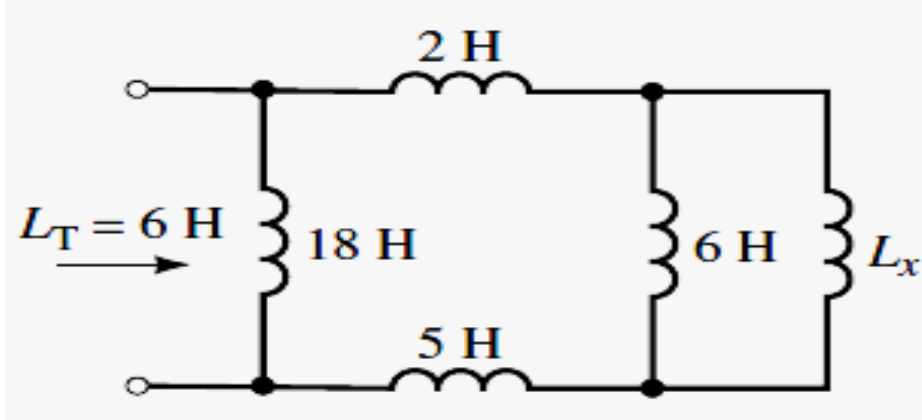
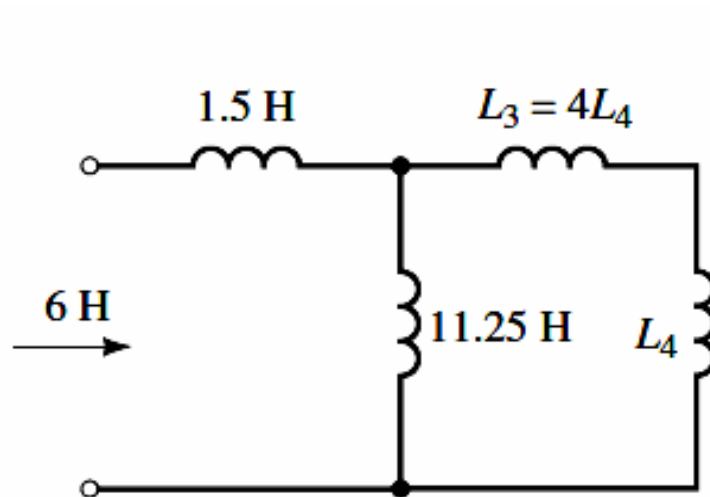
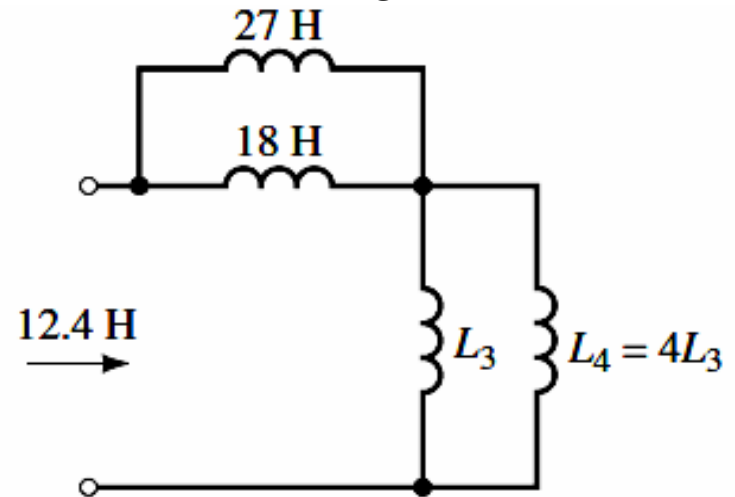


Figure 4.23

5) For the circuits of Figure 4.24, determine L_3 and L_4 .



(a)



(b)

Figure 4.24